Practical No. 2

Intents, Events and Listeners

1. Develop an Android application implementing Explicit and Implicit Intents.
2. Develop an Android application demonstrating Event Handling using Listeners
3. Develop an Android application using Adapters to populate Spinner or ListView dynamically.

AIM: To study and implement Programs on Intents, Events, Listeners and Adapters

- Explicit Intent
- Button Click Event
- Long Click Listener
- ListView with ArrayAdapter
- Toast Messages

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest package="com.example.intentdemo"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
        <activity android:name=".SecondActivity"></activity>
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
```

```
android:layout_width="match_parent"

   android:layout_height="match_parent">
   <LinearLayout
      android:orientation="vertical"
      android:padding="20dp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content">
      <TextView
         android:text="Intents & Events Demo"
         android:textSize="22sp"
         android:textStyle="bold"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"/>
      <!-- Intent Button -->
      <Button
         android:id="@+id/btnIntent"
         android:text="Open Second Activity"
         android:layout_marginTop="20dp"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"/>
      <!-- Toast Button -->
      <Button
         android:id="@+id/btnToast"
         android:text="Show Toast Message"
         android:layout_marginTop="10dp"
         android:layout_width="match_parent"
         android:layout_height="wrap_content"/>
      <!-- Long Click Button -->
      <Button
         android:id="@+id/btnLongClick"
```

```
            android:text="Long Press Me"
            android:layout_marginTop="10dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
        <!-- ListView -->
        <ListView
            android:id="@+id/listView"
            android:layout_marginTop="20dp"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
    </LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.intentdemo;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    Button btnIntent, btnToast, btnLongClick;
    ListView listView;
    String[] items = {"Item 1", "Item 2", "Item 3", "Item 4", "Item 5"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_main);
btnIntent = findViewById(R.id.btnIntent);
btnToast = findViewById(R.id.btnToast);
btnLongClick = findViewById(R.id.btnLongClick);
listView = findViewById(R.id.listView);
// Explicit Intent
btnIntent.setOnClickListener(v -> {
   Intent intent = new Intent(MainActivity.this, SecondActivity.class);
   startActivity(intent);
});
// Click Event Listener
btnToast.setOnClickListener(v ->
       Toast.makeText(MainActivity.this,
             "Button Clicked!",
             Toast.LENGTH_SHORT).show());
// Long Click Listener
btnLongClick.setOnLongClickListener(v -> {
   Toast.makeText(MainActivity.this,
         "Long Press Detected!",
         Toast.LENGTH_SHORT).show();
   return true;
});
// Adapter Example
ArrayAdapter<String> adapter =
       new ArrayAdapter<>(this,
             android.R.layout.simple_list_item_1,
             items);

listView.setAdapter(adapter);
```

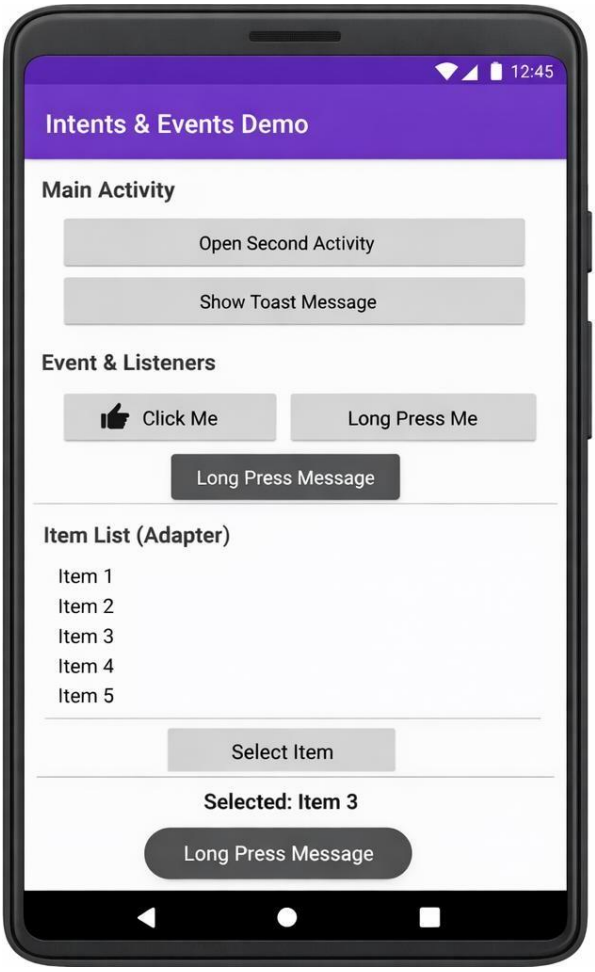
```
        // List Item Click Event
        listView.setOnItemClickListener((parent, view, position, id) ->
                Toast.makeText(MainActivity.this,
                        "Selected: " + items[position],
                        Toast.LENGTH_SHORT).show()) }`}
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:text="Welcome to Second Activity!"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

SecondActivity.java

```
package com.example.intentdemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```


12:45
Intents & Events Demo
Main Activity
Open Second Activity
Show Toast Message
Event & Listeners
Click Me
Long Press Me
Long Press Message
Item List (Adapter)
Item 1
Item 2
Item 3
Item 4
Item 5
Select Item
Selected: Item 3
Long Press Message


12:45
Welcome to Second Activity!